Hooke's Law

Aim: to observe Hooke's Law in different materials.

Equipment

- Retort stand
- Clamp
- Boss
- G clamp
- Metre ruler
- Mass hanger and 100g masses
- Spring
- Elastic band

Safety Information

Masses – risk of injury

Ensure the equipment is well secured on the desk.

Method

Step 1: Set up the retort stand, boss, clamp and spring as shown in the diagram.

Step 2: Fully secure the base of the retort stand to the desk and check it is firmly secure before continuing.

Step 3: Use the metre ruler to measure and record the original length of the spring (at 0g mass).

Step 4: Attach the mass hanger (with no masses on) to the spring. (The hanger should have a mass of 100g – check with your teacher if you are unsure).

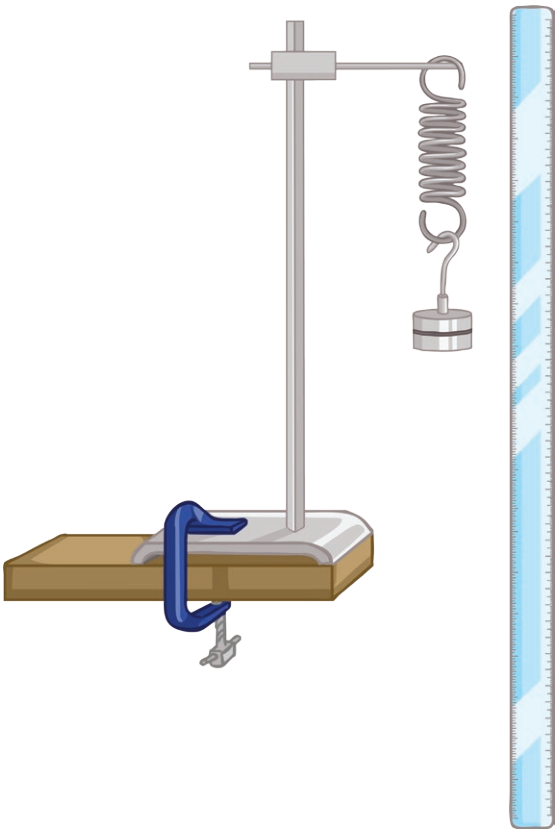
Step 5: Measure the new length of the spring and record.

Step 6: Place a 100g mass onto the mass hanger. Measure and record the new length of the spring.

Step 7: Continue to repeat step 6 until you have added 1000g (the hanger + 9 masses).

Step 8: Carefully unload the spring and remove from the apparatus.

Step 9: Attach the elastic band to the clamp in place of the spring and repeat the investigation.



Results Table

Make sure you write the correct units for each heading.

Mass (g)	Spring length (cm)	Spring extension (cm)	Elastic band length (cm)	Elastic band extension (cm)
0				
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

Hooke's Law states that an object will be stretched or squashed to an extension (e) which is directly proportional to the force (F) applied. Hooke's Law only applies within the elastic limit of the object.

$$F = k \times e$$

The spring constant (k) varies for different materials. If a spring is stiff (not easily stretched), it will have a greater value of k.

Questions

1. Calculate the extension of the spring and the elastic band and complete the table. To do this you need to subtract the original length of the spring (at 0g mass added) from the length of the spring. e.g. if the spring measured 6cm before any masses were added, and then measured 13cm with 100g added. The extension at 100g is 13 – 6 = 7cm.
2. Plot a graph to show extension against force for the spring and the elastic band. The extension should be plotted on the x-axis, in metres, and the force on the y-axis, in Newtons.
   To convert cm to m you will need to divide by 100. 100cm = 1m.
   100g = 1N.
3. Look at your graph. Does the spring obey Hooke's Law? Explain how you know.
4. Look at your graph. Does the elastic band obey Hooke's Law? Explain how you know.
5. Which part of the graph represents the spring constant?
6. Which material, the spring or the elastic band, has the highest spring constant? Explain how you know.

Answers

1. Check the student's individual results.
2. Check the student's individual results.
3. The spring should obey Hooke's Law and this will be evident by a directly proportional relationship between the extension and force. A straight-line graph should have been drawn. Depending on the type of spring used and its elastic limit, the graph may not be linear to the 10N force and the spring may have deformed during the practical investigation.
4. The elastic band should obey Hooke's Law and this will be evident by a directly proportional relationship between the extension and force. A straight-line graph should have been drawn.
5. The gradient of the line.
6. This will likely be the spring, and should be evident by a steeper gradient on the graph. (Or by a higher spring constant value if your students have calculated the spring constant for each material).